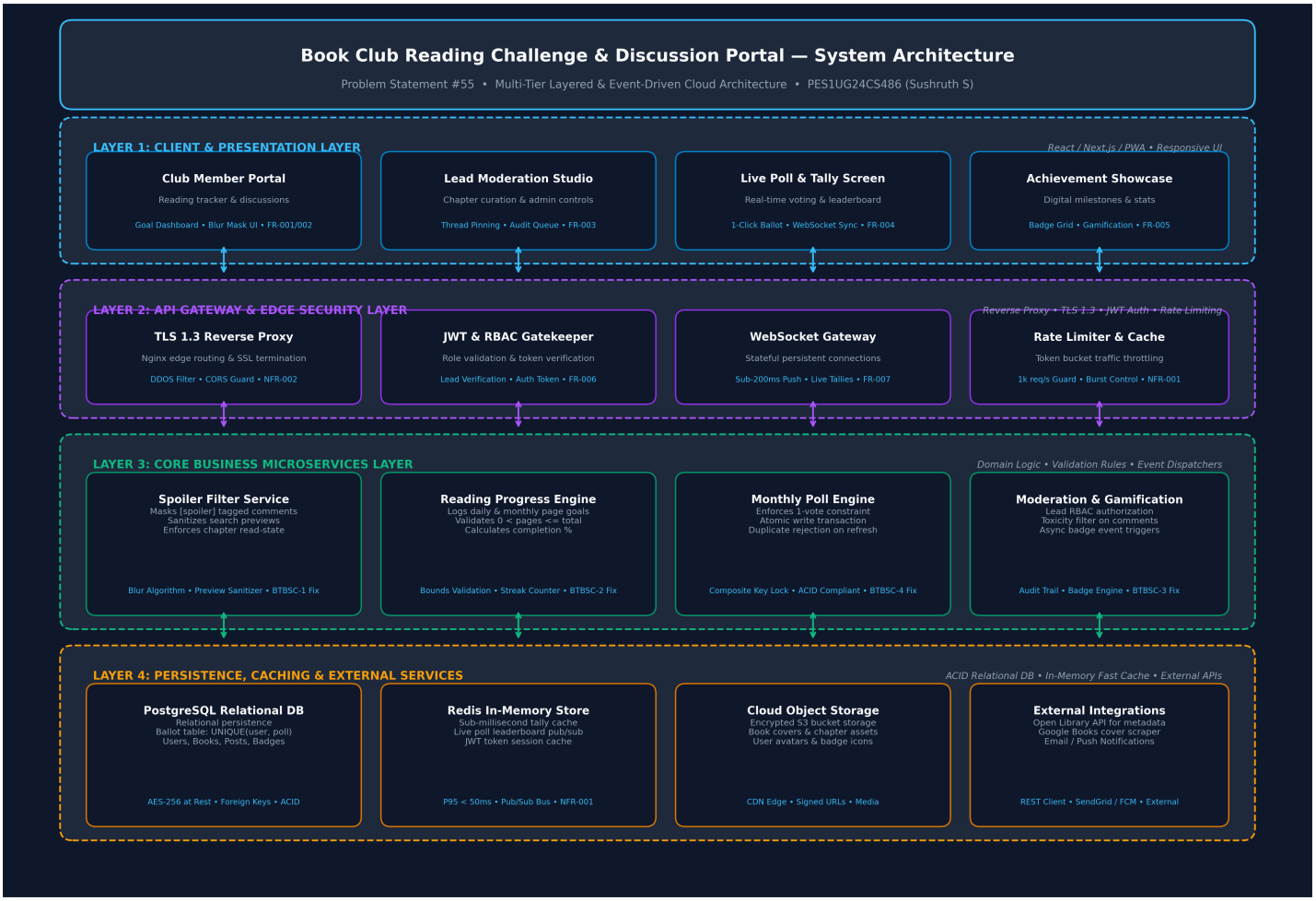
PES University — Department of Computer Science & Engineering

Lab 3: System Architecture & Component Specification

Problem Statement #55: Book Club Reading Challenge & Discussion Portal | **Domain:** Media, Events & Community
Student: Sushruth S | **SRN:** PES1UG24CS486 | **Section:** 4-H

1. System Architecture Overview & Diagram

The system is engineered as a **4-Tier Layered Architecture with Event-Driven Extensions**. It cleanly decouples user experience, edge security, business microservices, and persistence caching to meet strict performance and integrity goals.



2. Architectural Style Selection & Justification

• **Layered Multi-Tier Separation:** Isolates client views from relational storage, preventing direct database access and enforcing role-based authentication at the API Gateway level.
• **Sub-200ms Latency (NFR-001):** Offloads live monthly voting tallies from relational tables into an in-memory Redis cache paired with WebSockets, delivering real-time leaderboard updates without polling overhead.
• **ACID Transactional Integrity (NFR-004):** Enforces unique compound indexing on (user_id, poll_id) in PostgreSQL to eradicate duplicate voting attempts (remediating defect **BTBSC-4**).
• **Role-Based Access Control (FR-003):** Gateway interceptors validate Discussion Lead credentials cryptographically before requests enter the moderation service (remediating defect **BTBSC-3**).
• **Spoiler Masking & Sanitization (FR-001):** Server-side pre-sanitization obscures spoiler text in search previews while client-side blur masks protect in-thread discussions (remediating defect **BTBSC-1**).

3. Component Model & Subsystem Responsibilities

Component / Layer	Key Responsibilities	Interfaces & Protocols	Requirements / Bugs Addressed
Presentation Layer	Web/mobile interfaces, reading goal dashboard, blurred spoiler comments, live poll cards, achievement showcases.	HTTP/2, WebSocket Client, REST JSON	FR-001, FR-002, FR-004, FR-005, NFR-003
API Gateway & Edge	TLS 1.3 termination, JWT authentication, RBAC permission checks, token bucket rate limiting (1,000 req/s), WebSocket push.	Nginx / Envoy, WSS, REST reverse proxy	FR-006, FR-007, NFR-001, NFR-002, BTBSC-3
Spoiler Filter Service	Parses [spoiler] tags, renders CSS blur masks, unmasks on click or read-state, and sanitizes search index snippets.	REST API: /api/comments, /api/search	FR-001, BTBSC-1 Fix
Reading Progress Engine	Logs daily/monthly page progress, enforces bounds ($0 < ext{pages} \le ext{total}$), computes completion percentage, tracks streaks.	REST API: /api/users/{id}/progress	FR-002, BTBSC-2 Fix
Monthly Poll Engine	Conducts democratic monthly book voting with atomic transactions, composite unique constraint checks, and duplicate rejection.	REST API: /api/polls/{id}/vote	FR-004, NFR-004, BTBSC-4 Fix
Persistence & Cache Layer	PostgreSQL relational store with AES-256 encryption at rest; Redis in-memory cache for live tallies and sessions; S3 object storage.	PostgreSQL Wire Protocol, Redis RESP, S3 API	NFR-001, NFR-002, NFR-004